

Task: Given a sorted array of distinct integers `nums` and an integer `target`, return its index, or -1 if it is not present. (LeetCode 704)

Response:

```
def search(nums, target):  
    lo, hi = 0, len(nums) - 1  
    while lo < hi: # bug: should be lo <= hi  
        mid = (lo + hi) // 2  
        if nums[mid] == target: return mid  
        elif nums[mid] < target: lo = mid + 1  
        else: hi = mid - 1  
    return -1
```



Fail to repair



DrHall Mutation (Question mutation): In an ascending list of unique numbers, find the position of a given value, returning -1 when the value does not appear.



MutRepair Mutation: (Successful repair hallucination)

```
def search(nums, target):  
    i = bisect_left(nums, target)  
    if i < len(nums) and nums[i] == target:  
        return i  
    return -1
```